

LINEAR ALGEBRA

AND

CALCULUS

(LACE)

CO-1:- Linear Algebra

- 1) Evaluating Rank of matrix by reducing to Echelon Form.
- 2) Solution of system of equation.
 - (i) Gauss Elimination method
 - (ii) Lower & Upper

3) Quadratic form and it's canonical form

CO-2:- Differential Calculus

- 1) Partial Derivative
- 2) Evaluation of Jacobian
- 3) ~~Evaluation of~~ Total Derivatives and it's applications.
- 4) Chain Rule

5) Taylor series expansion for function with two variables.

6.) Evaluation of Maxima and minima for function with two variables or three variables.

7.) Lagrange's method.

8.) solution of second order ordinary differential ~~Expansion~~ Equation.

Calculus

Evaluation of Rank of a Matrix by reducing to Echelon form.

Rank of a Matrix:-

The rank of the matrix A' of order $m \times n$ is r .

- 1) At least one minor of order of $A \neq 0$.
- 2) All minors of order $> r$ or of A must be '0'.

Note:- The rank of null matrix is defined as '0'.

* The rank of the matrix A' is denoted by $\rho(A)$.

* If A is $m \times n$ matrix, $\rho(A) \leq \min(m, n)$

Elementary Row Transformation;

Echelon form

* A matrix is said to be Echelon form if the min number of zero elements before the non-zero element in each row must be Increased or ~~decreased~~ same.

* Note:- If the matrix A' is in echelon form, the rank of the matrix $\rho(A)$ is defined as the number of non-zero rows of the matrix.

Session - 2

1) Find the rank of the matrix by reducing to echelon form.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

upper
triangular
matrix

Sol:- As the given matrix is a square matrix the upper triangular matrix have to be zeros then it gets reduced to echelon form.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \\ 0 & -1 & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$\Rightarrow$ 3 non-zero rows

$$\text{Rank} \Rightarrow \rho(A) = 3$$

② Evaluate rank of the matrix. 1/1/25

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

Sol:-

$$\begin{aligned} R_2 &\leftrightarrow R_1 \\ R_2 &\rightarrow R_2 - R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \\ R_4 &\rightarrow R_4 - 6R_1 \end{aligned}$$

$$\begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 3 & +3 & +7 \\ 0 & 4 & 9 & +7 \\ 0 & 9 & 12 & +7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_3$$

$$\begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & +1 & -6 & -3 \\ 0 & 4 & 9 & +7 \\ 0 & 9 & 12 & +7 \end{bmatrix}$$

$$\begin{aligned} R_3 &\rightarrow R_3 - 4R_2 \\ R_4 &\rightarrow R_4 - 9R_2 \end{aligned}$$

$$\begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 66 & 44 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 2R_3$$

$$\begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

diagonal elements ↑

* principal may or may not be zeros

$$\text{Rank } P(A) = 3$$

Note:- Consider $a_1x + b_1y + c_1z = d_1$
 $a_2x + b_2y + c_2z = d_2$
 $a_3x + b_3y + c_3z = d_3$

can be written in the form

$$AX = B$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}_{3 \times 3} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1} = B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}_{3 \times 1}$$

(co-efficient matrix) $3 \times 3 \times 3 \times 1 = 3 \times 1$

Augmented matrix

$$= [A \quad B]$$

$$= \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$$

③ Determine rank of the coefficient matrix for the following system.

$$2x + 3y + z = 6$$

$$4x + 5y + z = 10$$

$$x + y + 3z = 5$$

$$\begin{bmatrix} 2 & 3 & 1 \\ 4 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$$

Co-efficient matrix

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3$$

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 4 & 5 & 1 \\ 2 & 3 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 4R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -11 \\ 0 & 1 & -5 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -11 \\ 0 & 0 & 6 \end{bmatrix} \quad 3 \times 3$$

$$\text{Rank } \rho(A) = 3$$

- ④ Two containers contain a water of diff temperatures. If we mix 240g of water from the first container with 260g of water from the second container the resulting water temp will be 52°C . If we mix 180g from 1st container with 120g of water from 2nd container the resulting temp is 46°C .

model this phenomena as system of equations and hence find rank of the coefficient matrix.

Sol:- Take x° & y° are temperatures of water in 1st & 2nd containers respectively.

By the given data,

$$24\phi x + 26\phi y = 52.50\phi$$

$$18\phi x + 12\phi y = 46.30\phi$$

$$\begin{cases} 24x + 26y = 2600 \\ 18x + 12y = 1360 \end{cases} \rightarrow \text{① system of equations}$$

Co-efficient matrix
$$= \begin{bmatrix} 24 & 26 \\ 18 & 12 \end{bmatrix}$$

$$R_2 \rightarrow 24R_2 - 18R_1$$

$$A = \begin{bmatrix} 24 & 26 \\ 0 & -180 \end{bmatrix}$$

$$R(A) = 2$$

⑧ Note:- If 3 vectors v_1, v_2 & v_3 are coplanar, then the rank of the matrix A formed by v_1, v_2 & v_3 is < 3 .

$\rightarrow$ IF v_1, v_2 & v_3 are non-coplanar then $R(A) = 3$.

* IF $|A| = 0$ (coplanar); $|A| \neq 0$ (non-coplanar)

⑤ Verify the given Vectors $[1 \ -2 \ 3]$,
 $[-2 \ 4 \ 6]$, $[3 \ -6 \ 9]$.

Sol:- Take $A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & 6 \\ 3 & -6 & 9 \end{bmatrix}$

$$\begin{aligned} R_2 &\rightarrow R_2 + 2R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 0 & 12 \\ 0 & 0 & 0 \end{bmatrix}$$

$R(A) = 2$, As $2 < 3$ the vectors are coplanar.

⑥ Determine rank of the matrix

$$A = \begin{bmatrix} 0 & 4 & 3 & 2 & 1 \\ 0 & -2 & -3 & -1 & 4 \\ 0 & 6 & 7 & 2 & 9 \\ 0 & 8 & 3 & 6 & 6 \end{bmatrix}$$

$$\sim R_2 \leftrightarrow R_1$$

$$A = \begin{bmatrix} 0 & -2 & -3 & -1 & 4 \\ 0 & 4 & 3 & 2 & 1 \\ 0 & 6 & 7 & 2 & 9 \\ 0 & 8 & 3 & 6 & 6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_1$$

$$R_3 \rightarrow R_3 + 3R_1$$

$$R_4 \rightarrow R_4 + 4R_1$$

$$A = \begin{bmatrix} 0 & -2 & -3 & -1 & 4 \\ 0 & 0 & -3 & 0 & 9 \\ 0 & 0 & -2 & -1 & 3 \\ 0 & 0 & -9 & 2 & 22 \end{bmatrix}$$

$$R_3 \rightarrow +3R_3 - (+2R_2)$$

$$R_3 \rightarrow +3R_3 + 2R_2$$

$$A = \begin{bmatrix} 0 & -2 & -3 & -1 & 4 \\ 0 & 0 & -3 & 0 & 9 \\ 0 & 0 & -2 & -1 & -3 \\ 0 & 0 & -9 & 12 & 22 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

matrix A is 4×5 matrix. $\therefore (A) \neq$

invertible. $\therefore$ it is not invertible.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = A$$

$1A \leftrightarrow 5A$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = A$$

$1A \leftrightarrow 2A$

$1A \leftrightarrow 3A$

Solution of system of equations 12/1/25

Methods:-

① Gauss-Elimination method.

Process:-

- Reduce 'A' into Upper Triangular Matrix by using row transformation.
- By taking back substitutions we get z, y, x values respectively.

② Solve the system of equations by

$$2x + y + z = 10$$

$$3x + 2y + 3z = 18$$

$$x + 4y + 9z = 16$$

method.

A

x

= B

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \\ 16 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$$

$\sim R_1 \leftrightarrow R_3$

$$\begin{bmatrix} 1 & 4 & 9 \\ 3 & 2 & 3 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ 18 \\ 10 \end{bmatrix}$$

3×3

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 4 & 9 \\ 0 & -10 & -24 \\ 0 & -7 & -17 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ -30 \\ -22 \end{bmatrix}$$

$$R_3 \rightarrow 10R_3 - 7R_2$$

$$\begin{bmatrix} 1 & 4 & 9 \\ 0 & -10 & -24 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ -30 \\ -10 \end{bmatrix}$$

$$x + 4y + 9z = 16$$

$$-10y - 24z = -30$$

$$-2z = -10$$

$$\boxed{z = 5}$$

$$-10y - 24(5) = -30$$

$$-10y - 120 = -30$$

$$+10y = 90$$

$$y = \frac{90}{-10} = -9$$

$$\boxed{y = -9}$$

$$x + 4(-9) + 9(5) = 16$$

$$x - 36 + 45 = 16$$

$$x + 9 = 16$$

$$x = 16 - 9$$

$$\boxed{x = 7}$$

② A company can produce 3 types of cloths A, B & C. 3 types of wool are required they are red, green & blue wool. One unit of type 'A' cloth requires two yards of red wool, 8 yards of green & 1 yard of blue wool. For manufacturing 1 unit of type 'B' cloth requires 1 yard red, 3 yards green & 5 yards of blue wool. For manufacturing one unit of type 'C' cloth we require 6 yards of red, 2 yards of green & 1 yard of blue. Company has only a stock of 9 yards of red 13 yards of green & 7 yards blue. If total stock is used, then determine the number of units of type 'A', 'B' & 'C' cloths produced by company. By reducing the given phenomena as system of equations, by using Gauss - elimination method.

Sol:- Take no of units of types A, B and C cloths are x , y and z respectively

Given

	R	G	B	
A $\rightarrow$	2	8	1	$\Rightarrow$
B $\rightarrow$	1	3	5	
C $\rightarrow$	6	2	1	
	9	13	7	

$$\begin{aligned}
 2x + 8y + z &= 9 \\
 x + 3y + 5z &= 13 \\
 x + 5y + z &= 7
 \end{aligned}$$

$$A \begin{bmatrix} 2 & 1 & 6 \\ 8 & 3 & 2 \\ 1 & 5 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 13 \\ 7 \end{bmatrix}$$

$$\sim R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 5 & 1 \\ 8 & 3 & 2 \\ 2 & 1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 13 \\ 9 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 8R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 5 & 1 \\ 0 & -37 & -6 \\ 0 & -9 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ -43 \\ -5 \end{bmatrix}$$

$$R_3 \rightarrow 37R_3 - 9R_2$$

$$\begin{bmatrix} 1 & 5 & 1 \\ 0 & -37 & -6 \\ 0 & 0 & 202 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ -43 \\ 202 \end{bmatrix}$$

$$x + 5y + z = 7$$

$$-37y - 6z = -43$$

$$202z = 202$$

$$\boxed{z = 1}$$

$$-37y - 6(1) = -43$$

$$-37y = -37$$

$$\boxed{y = 1}$$

$$x + 5(1) + 1 = 7$$

$$x + 6 = 7$$

$$\boxed{x = 1}$$

No of units company
can produce 1 unit
of each type A, B's
'C' cloths

Solution of system of Equations by using L.V. Decomposition method

Note:- If the given square matrix 'A' is non-singular, then 'A' can be uniquely expressed as $A = LU$

$$A = LU = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

(L.T.M) (U.T.M)

$$\left. \begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned} \right\} \rightarrow \text{①}$$

$$AX = B$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

STEP 1:-

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

By composing we can find L and U matrices.

STEP II :-

w.k.t $AX=B$

$$A=LU$$

$$LUX=B \rightarrow \textcircled{2}$$

Take $U_{3 \times 3} X_{3 \times 1} = V_{3 \times 1}$

$$V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

from $\textcircled{2}$

$$LV=B \rightarrow \textcircled{3}$$

By simplification we get V

STEP III :-

from $\textcircled{3}$

$$UX=V$$

By simplification we get X

Q. Solve the system $3x + 2y + 7z = 4$
 $2x + 3y + z = 5$
 $3x + 4y + z = 7$ by
 using L.V. Decomposition method.

Sol:- $Ax = B$

$$A = \begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix} x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = B = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$LUX = B$$

Compare the two matrices.

Ist row
 $u_{11} = 3; u_{12} = 2; u_{13} = 7$

II row

$$l_{21} u_{11} = 2$$

$$l_{21} \cdot 3 = 2$$

$$l_{21} = \frac{2}{3}$$

$$l_{21} u_{12} + u_{22} = 3$$

$$\frac{2}{3} \times 2 + u_{22} = 3$$

$$\frac{4}{3} + u_{22} = 3$$

$$u_{22} = 3 - \frac{4}{3} = \frac{5}{3}$$

$$l_{21} u_{13} + u_{23} = 1$$

$$\frac{2}{3} \times 7 + u_{23} = 1$$

$$\frac{14}{3} + u_{23} = 1$$

$$u_{23} = 1 - \frac{14}{3}$$

$$u_{23} = \frac{-11}{3}$$

Row 3

$$l_{31}u_{12} + l_{32}u_{22} = 4$$

$$3 \times 2 + l_{32} \times \frac{5}{3} = 4$$

$$6 + l_{32} \times \frac{5}{3} = 4$$

$$l_{32} \times \frac{5}{3} = 4 - 6$$

$$l_{32} \times \frac{5}{3} = -2$$

$$l_{32} = -2 \times \frac{3}{5}$$

$$= -\frac{6}{5}$$

$$l_{31}u_{13} + l_{32}u_{23} = 7$$

$$3 \times 7 + \frac{5}{3} \times \left(-\frac{11}{3}\right) = 7$$

$$= 1$$

$$l_{33} = \frac{-66}{18} = -\frac{11}{3}$$

$$l_{33} = \frac{-22}{5} + u_{33}$$

$$l_{33} = \frac{-22}{5} + u_{33}$$

$$u_{33} = -6 + \frac{33}{5}$$

$$u_{33} = \frac{-8}{5}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1 & -6/5 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 3 & 2 & 7 \\ 0 & 5/3 & -11/3 \\ 0 & 0 & -8/5 \end{bmatrix}$$

$$LUX = B$$

$$UX = V$$

$$\begin{bmatrix} 3 & 2 & 7 \\ 0 & 5/3 & -11/3 \\ 0 & 0 & -8/5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$LV = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1 & 6/5 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ 2/3 v_1 + v_2 \\ v_1 + 6/5 v_2 + v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

$$v_1 = 4$$

$$\frac{2}{3} v_1 + v_2 = 5$$

$$\frac{2}{3} \times 4 + v_2 = 5$$

$$\frac{8}{3} + v_2 = 5$$

$$v_2 = 5 - \frac{8}{3}$$

$$v_2 = \frac{7}{3}$$

$$v = \begin{bmatrix} 4 \\ 7/3 \\ 15 \end{bmatrix}$$

$$\begin{aligned} v_1 + 6/5 v_2 + v_3 &= 7 \\ 4 + 6/5 \times \frac{7}{3} + v_3 &= 7 \\ 4 + \frac{42}{5} + v_3 &= 7 \end{aligned}$$

$$\frac{42}{5} + v_3 = 3$$

$$v_3 = 3 - \frac{42}{5}$$

$$v_3 = \frac{15 - 42}{5}$$

$$v_3 = \frac{-27}{5} = -\frac{27}{5}$$

$$UX = V$$

$$\begin{bmatrix} 3 & 2 & 7 \\ 0 & 5/3 & -11/3 \\ 0 & 0 & -8/5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 7/3 \\ 1/5 \end{bmatrix}$$

$$UX = \begin{bmatrix} 3x + 2y + 7z \\ 5/3 y - 11/3 z \\ -8/5 z \end{bmatrix} = \begin{bmatrix} 4 \\ 7/3 \\ 1/5 \end{bmatrix}$$

$$-8/5 z = 1/5$$

$$z = \frac{1}{8} \times \frac{5}{8}$$

$$z = -\frac{1}{8}$$

$$\frac{5}{3} y - \frac{11}{3} z = \frac{7}{3}$$

$$\frac{5}{3} y - \frac{11}{3} \left(-\frac{1}{8}\right) = \frac{7}{3}$$

$$\frac{5}{3} y + \frac{11}{24} = \frac{7}{3}$$

$$\frac{5}{3} y = \frac{7}{3} - \frac{11}{24}$$

$$= \frac{56 - 11}{24}$$

$$\frac{5}{3} y = \frac{45}{24}$$

$$y = \frac{45}{24} \times \frac{3}{5}$$

$$y = \frac{9}{8}$$

$$3x + 2y + 7z = 4$$

$$3x + 2\left(\frac{9}{8}\right) + 7\left(-\frac{1}{8}\right) = 4$$

$$3x + \frac{9}{4} - \frac{7}{8} = 4$$

$$3x + \frac{18 - 7}{8} = 4$$

$$3x + \frac{11}{8} = 4$$

$$3x = 4 - \frac{11}{8}$$

$$3x = \frac{32 - 11}{8}$$

$$3x = \frac{21}{8}$$

$$x = \frac{21}{8} \times \frac{1}{3}$$

$$x = \frac{7}{8}$$

$$X = \begin{bmatrix} 7/8 \\ 9/8 \\ -1/8 \end{bmatrix}$$

② Print India Ltd, wishes to produce three types of sovereigns 'A', 'B', 'C' to manufacture a unit of type 'A' sovereign requires 2 minutes on machine 1, 1 min on machine 2 & 2 min on machine 3. A unit of Type 'B' sovereign requires 1 min on machine 1, 3 min on machine 2 & 1 min on machine 3. One unit of Type 'C' sovereign requires 1 min on machine 1 & 2 min on machine 2 & 2 min on machine 3. There are 180 min available on machine 1 & 300 min available on machine 2 & 240 min available on machine 3.

How many sovereigns of each type that company will produce by using an available time by applying L.V. Decomposition method.

Sol:- Take x, y & z are the type A, B, C respectively.

	m_1	m_2	m_3
A	2m	1m	2m
B	1m	3m	1m
C	1m	2m	2m
	180	300	240

$$\Rightarrow \begin{cases} x + y + z = 180 \\ x + 3y + z = 300 \\ 2x + y + 2z = 240 \end{cases} \rightarrow (1)$$

$$A X = B$$

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 180 \\ 300 \\ 240 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 2, u_{12} = 1, u_{13} = 1$$

Ist row

$$\begin{array}{l|l} l_{21}u_{11} = 1 & l_{21}u_{12} + u_{22} = 3 \\ l_{21} \cdot 2 = 1 & \frac{1}{2} \times 1 + u_{22} = 3 \\ l_{21} = \frac{1}{2} & \frac{1}{2} + u_{22} = 3 \\ & u_{22} = 3 - \frac{1}{2} \\ & u_{22} = \frac{5}{2} \end{array} \quad \begin{array}{l} l_{21}u_{13} + u_{23} = 2 \\ \frac{1}{2} \times 1 + u_{23} = 2 \\ u_{23} = 2 - \frac{1}{2} \\ u_{23} = \frac{3}{2} \end{array}$$

IInd row:-

$$\begin{array}{l|l} l_{31}u_{11} = 2 & l_{31}u_{12} + l_{32}u_{22} = 1 \\ l_{31} \cdot 2 = 2 & 1 \times 1 + l_{32} \cdot \frac{5}{2} = 1 \\ \boxed{l_{31} = 1} & l_{32} \times \frac{5}{2} = 0 \\ & \boxed{l_{32} = 0} \end{array} \quad \begin{array}{l} l_{31}u_{13} + l_{32}u_{23} + u_{33} = 2 \\ 1 \times 1 + 0 + u_{33} = 2 \\ \boxed{u_{33} = 1} \end{array}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & \frac{5}{2} & \frac{3}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

$$L V = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 180 \\ 300 \\ 240 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ \frac{1}{2}v_1 + v_2 \\ v_1 + v_3 \end{bmatrix} = \begin{bmatrix} 180 \\ 300 \\ 240 \end{bmatrix}$$

$$v_1 = 180 \quad \begin{array}{l} \frac{1}{2}v_1 + v_2 = 300 \\ \frac{1}{2} \times 180 + v_2 = 300 \\ 90 + v_2 = 300 \\ \boxed{v_2 = 210} \end{array}$$

$$U X = V$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & \frac{5}{2} & \frac{3}{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 180 \\ 300 \\ 240 \end{bmatrix}$$

$$\begin{bmatrix} 2x + y + z \\ \frac{5}{2}y + \frac{3}{2}z \\ z \end{bmatrix} = \begin{bmatrix} 180 \\ 300 \\ 240 \end{bmatrix}$$

$$\boxed{z = 240} \quad \begin{array}{l} \frac{5}{2}y + \frac{3}{2}z = 300 \\ \frac{5}{2}y + \frac{3}{2} \times 240 = 300 \\ \frac{5}{2}y + 360 = 300 \\ \frac{5}{2}y = 300 - 360 \\ \frac{5}{2}y = -60 \\ y = -24 \end{array}$$

$$1 \quad V = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 180 \\ 300 \\ 240 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ \frac{1}{2}v_1 + v_2 \\ v_1 + v_3 \end{bmatrix} = \begin{bmatrix} 180 \\ 300 \\ 240 \end{bmatrix}$$

$$v_1 = 180 \quad \left| \begin{array}{l} \frac{1}{2}v_1 + v_2 = 300 \\ \frac{1}{2} \cdot 180 + v_2 = 300 \\ 90 + v_2 = 300 \\ v_2 = 210 \end{array} \right.$$

$$\begin{array}{l} v_1 + v_3 = 240 \\ 180 + v_3 = 240 \\ v_3 = 60 \end{array}$$

$$Ux = V$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 5/2 & 3/2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 180 \\ 210 \\ 60 \end{bmatrix}$$

$$\begin{bmatrix} 2x + y + z \\ 5/2 y + 3/2 z \\ z \end{bmatrix} = \begin{bmatrix} 180 \\ 210 \\ 60 \end{bmatrix}$$

$$\boxed{z = 60} \quad \left| \begin{array}{l} \frac{5}{2}y + \frac{3}{2}z = 210 \\ \frac{5}{2}y + \frac{3}{2} \cdot 60 = 210 \\ \frac{5}{2}y + 90 = 210 \end{array} \right.$$

$$\begin{array}{l} \frac{5}{2}y = 120 \\ y = 48 \end{array}$$

$$2x + 48 + 60 = 180$$

$$2x + 108 = 180$$

$$2x = 72$$

$$\boxed{x = 36}$$

$$\therefore x = \begin{bmatrix} 36 \\ 48 \\ 60 \end{bmatrix} \rightarrow \begin{array}{l} \text{No of type 'A' sovereigns} \\ \text{No of type 'B' sovereigns} \\ \text{No of type 'C' sovereigns} \end{array}$$

09/01/25

Eigen Values and Eigen Vectors

① $|A - \lambda I|$ is characteristic polynomial of 'A'.

② $|A - \lambda I| = 0$ is characteristic equation of 'A'.

③ Roots of characteristic equation are called Eigen values.

* Where 'I' is Identity matrix.

* When 'A' is 2×2 matrix to find the characteristic equation: ~~$\lambda^2 - 5\lambda - \dots$~~

$$\boxed{\lambda^2 - (\text{Trace of } A)\lambda - |A| = 0}$$

* When 'A' is 3×3 matrix to find the characteristic equation =

$$\boxed{\lambda^3 - (\text{Trace of } A)\lambda^2 + \left(\sum \text{minors of principal diagonal elements}\right)\lambda - |A| = 0}$$

Sum of minors
of principal
diagonal elements

Properties of Eigen Values:-

- ① The sum of Eigen values of square matrix $A' = \text{Trace of } (A)$
- ② The product of Eigen values of square matrix $A' = |A|$
- ③ The Eigen values of Triangular matrix are principal diagonal elements.
- ④ If $\lambda_1, \lambda_2, \lambda_3$ are eigen values of $(A)_{3 \times 3}$
Then:
i) The eigen values of A^2 are $\lambda_1^2, \lambda_2^2, \lambda_3^2$
ii) The eigen values of A^3 are $\lambda_1^3, \lambda_2^3, \lambda_3^3$
iii) The eigen value of A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}$

Session - 5

- ① Determine Eigen values of $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}_{2 \times 2}$

$$\Rightarrow \lambda^2 - 7\lambda + 6 = 0 \quad |A - \lambda I| = 0$$
$$\boxed{\lambda^2 - 7\lambda + 6 = 0}$$
$$A - \lambda I = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda^2 - 6\lambda - \lambda + 6 = 0 \quad \text{or} \quad \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\lambda(\lambda - 6) - 1(\lambda - 6)$$
$$(\lambda - 1)(\lambda - 6)$$
$$= \begin{bmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{bmatrix}$$

$$\therefore \boxed{\lambda = 1, 6}$$

Note:- Consider the system $\frac{dx}{dt} = Ax$ with matrix A . Then system (1) is stable if all Eigen values of A are negative.

* Consider the given system (2) is unstable in all other cases.

Eg: (i) If $1 \pm 2i$ are eigen values at A .

* If eigen values are complex then check the real part is +ve or -ve. If real part is +ve or zero, it is unstable, if -ve it is stable.

(i) $1 \pm 2i$ is +ve so it is unstable.

(ii) If $-2, -3, 1$ are eigen values at A .

* All the values should be -ve then only it is stable, otherwise it is unstable.

Eigen Vector

The non-zero solution of the system $(A - \lambda I)x = 0$ is called a eigen vector of A corresponding to characteristic root λ of A .

(2) Determine Eigen value & vector for the matrix.

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

$$\text{Sol: } A - \lambda I = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix}$$

(or)

The characteristic eqⁿ of $A_{3 \times 3}$ is
 $\lambda^3 - (\text{Trace of } A)\lambda^2 + \lambda \left[\begin{matrix} \text{sum of} \\ \text{minors of} \\ \text{diagonal} \\ \text{elements} \end{matrix} \right] - 1$

$$\text{minor of } 8 = \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} = 21 - 16 = 5$$

$$\text{minor of } 7 = \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} = 24 - 4 = 20$$

$$\text{minor of } 3 = \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix} = 56 - 36 = 20$$

$$\lambda^3 - 18\lambda^2 + 45\lambda - 0 = 0$$

$$\lambda^3 - 18\lambda^2 + 45\lambda = 0$$

$$\lambda(\lambda^2 - 18\lambda + 45) = 0$$

$$\lambda^2 - 18\lambda + 45 = 0$$

$$\lambda(\lambda - 15) - 3(\lambda - 15) = 0$$

$$(\lambda - 3)(\lambda - 15)$$

$$\lambda = 3, 15, 0$$

$$\begin{aligned} 6x - 6y + 2z &= 0 \\ -6x + 7y + 4z &= 0 \\ 2x - 4y + 3z &= 0 \end{aligned}$$

$$\begin{array}{ccc|c} x & y & z & \\ \hline -6 & 2 & 8 & -6 \\ 7 & -4 & -6 & 7 \end{array} \rightarrow \frac{x}{24-14} = \frac{y}{-12+32} = \frac{z}{56-36}$$

$$\frac{x}{10} = \frac{y}{20} = \frac{z}{20} = k \Rightarrow x = k, y = 2k, z = 2k$$

$$X = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

For $\lambda = 3$

$$(A - 3I)X = 0$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 5x - 6y + 2z &= 0 \\ -6x + 4y - 4z &= 0 \\ 2x - 4y &= 0 \end{aligned}$$

$$\begin{array}{ccc|c} x & y & z & \\ \hline -6 & 2 & 5 & -6 \\ 4 & -4 & -6 & 4 \end{array}$$

$$\frac{x}{24-8} = \frac{y}{-12+20} = \frac{z}{20-36}$$

$$\frac{x}{16} = \frac{y}{8} = \frac{z}{-16} = k$$

$$\frac{x}{9} = \frac{y}{1} = \frac{z}{-2} = k$$

$$X = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$$x = 2k, y = k, z = -2k$$

for $A=19$

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-7x - 6y + 2z = 0$$

$$= 0$$

$$-6x - 8y - 4z = 0$$

$$= 0$$

$$2x - 4y - 12z = 0$$

$$\frac{x}{24+16} = \frac{y}{-12+28} = \frac{z}{56-36}$$

$$\frac{x}{40} = \frac{y}{16} = \frac{z}{20} = k$$

$$\frac{x}{2} = \frac{y}{-2} = \frac{z}{1} = k$$

$$X = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

Q) Determine Eigen values of the matrix A .

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\lambda = 2, 3, 5 \rightarrow \text{Eigen values.}$$

$$\underline{\lambda = 2}$$

$$(A - 2I) = 0$$

$$\begin{bmatrix} 0 & 2 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

$$2y - z = 0$$

$$y + 4z = 0$$

$$3z = 0$$

$$\frac{x}{9} = \frac{y}{0} = \frac{z}{0} = k$$

$$x = 9k$$

$$y = 0$$

$$z = 0$$

$$x = \begin{bmatrix} 9 \\ 0 \\ 0 \end{bmatrix} \rightarrow \text{Eigen Vector}$$

$$\underline{\lambda = 3}$$

$$(A - 3I) = 0$$

$$\begin{bmatrix} -1 & 2 & -1 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{aligned} -x + 2y - z &= 0 \\ 4z &= 0 \\ 2z &= 0 \end{aligned}$$

$$\begin{array}{ccc|c} x & y & z & \\ \hline 8 & -1 & -1 & 2 \\ 0 & 4 & 0 & 0 \end{array} \quad \frac{x}{8} = \frac{y}{4} = \frac{z}{0} = k$$

$$x = 8k, y = 4k, z = 0$$

$$x = \begin{bmatrix} 8 \\ 4 \\ 0 \end{bmatrix} \rightarrow \text{Eigenvector}$$

$$\lambda = 5$$

$$(A - 5I) = 0$$

$$\begin{bmatrix} -3 & 2 & -1 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} -3x + 2y - z &= 0 \\ -2y + 4z &= 0 \end{aligned}$$

$$\begin{array}{ccc|c} x & y & z & \\ \hline 2 & -1 & -3 & 2 \\ -2 & 4 & 0 & -2 \end{array}$$

$$\frac{x}{8-2} = \frac{y}{12} = \frac{z}{6} = k$$

$$\frac{x}{6} = \frac{y}{12} = \frac{z}{6} = k$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{1} = k$$

$$x = k, y = 2k, z = k$$

$$x = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \rightarrow \text{eigen vector}$$

Q. If two eigen values of $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix}$ are 1 & 5. Then Find third eigen value of 'A' & also Find eigen values of A^4 , A^7 & A^{-1} .

Sol:- Sum of eigen values = Trace of A

$$\lambda_1 + \lambda_2 + \lambda_3 = 7$$

$$1 + 5 + \lambda_3 = 7$$

$$\boxed{\lambda_3 = 1}$$

$$A^4 \Rightarrow \lambda_1^4 = 1^4 = 1 \quad \lambda_2^4 = 5^4 = 625 \quad \lambda_3^4 = 1^4 = 1$$

$$A^7 \Rightarrow \lambda_1^7 = 1^7 = 1 \quad \lambda_2^7 = 5^7 = 78125 \quad \lambda_3^7 = 1^7 = 1$$

$$A^{-1} = \frac{1}{\lambda_1} \mid \frac{1}{\lambda_2} \mid \frac{1}{\lambda_3}$$

$$= \frac{1}{1} \mid \frac{1}{5} \mid \frac{1}{1}$$

$$A^T = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix}$$

~~$$\lambda_1 = 2, \lambda_2 = 2$$~~

$$\lambda_1 = 1, \lambda_2 = 5, \lambda_3 = ?$$

$$1 + 5 + \lambda_3 = 7$$

$$\boxed{\lambda_3 = 1}$$

Diagonalization of a matrix

* Consider a square matrix of order 3×3 .
To verify 'A' is diagonalisable or not, we consider the following steps.

Step 1:- Find all eigen values of 'A' & takes $\lambda_1, \lambda_2, \lambda_3$.

Step 2:- Find eigen vectors of 'A' corresponding to each eigen value $\lambda_1, \lambda_2, \lambda_3$ and take as x_1, x_2 & x_3 respectively.

Step 3:- Consider the modal matrix 'P' as as $P = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}$

Step 4:-

① If $|P| \neq 0$, then 'A' is not diagonalisable.

Note:- $D = P^{-1}AP$

$$D^2 = P^{-1}A^2P$$

$$A^n = PP^{-1}D^nP^{-1}P = PD^nP^{-1}$$

Session - 6

① Find the system is stable or not.

Sol:- $(A - \lambda I) = 0$

$$\lambda^2 - (\text{Trace of } A)\lambda + |A| = 0$$

$$\lambda^2 - 5\lambda + \overset{-6}{-6} = 0$$

$$\lambda^2 - 6\lambda + \lambda - 6 = 0$$

$$\lambda(\lambda - 6) + 1(\lambda - 6) = 0$$

$$A = \begin{bmatrix} 1 & 2 \\ -5 & 4 \end{bmatrix}$$

$$\textcircled{2} A = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

$$\lambda^3 - 6\lambda^2 + (0 + 0 + 0)\lambda - |A|$$

$$|A| = 1(4 - 4) - (-2)(-2 - (-2)) + 1(4 - 4)$$

$$= 0 + 0 + 0$$

$$\lambda^3 - 6\lambda^2 + 0\lambda - 0$$

$$\boxed{\lambda^3 - 6\lambda^2 = 0}$$

$$\lambda^2(\lambda - 6) = 0$$

$$\lambda = 0, \lambda = 0, \lambda = 6$$

$\therefore$ System is unstable.

Session-7

\textcircled{1} Verify $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ is diagonalisable or not.
If diagonalisable find A^n .

Sol:-

$$\lambda^2 - (\text{Trace of } A)\lambda + |A|$$

$$= \lambda^2 - 4\lambda - 5$$

$$\lambda^2 - 5\lambda + \lambda - 5$$

$$\lambda(\lambda - 5) + 1(\lambda - 5)$$

$$\boxed{\lambda = -1, 5}$$

$$\underline{\lambda = -1}$$

$$(A - \lambda I) = (A - (-1)I)$$

$$(A + I)$$

$$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} 2x + 2y = 0 \\ 4x + 4y = 0 \end{cases} \Rightarrow x + y = 0$$

Take $y = k \neq 0$

$$x_1 = k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

for $k = 1$

$$x_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\underline{\lambda = 5}$$

$$(A - \lambda I) = (A - 5I)$$

$$\begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} -4x + 2y = 0 \\ 4x - 2y = 0 \end{cases} \Rightarrow 2x = y$$

$y = k \neq 0$

$x = \frac{k}{2}$

$$x_2 = \begin{bmatrix} k/2 \\ k \end{bmatrix} = k \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

for $k = 2$

$$x_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$P = [x_1 \ x_2]$$

$$= \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$|P| = -2 - 1$$

$$\boxed{|P| = -3}$$

$\therefore 'A'$ is diagonalisable.

$$D = P^{-1} A P$$

$$= \frac{1}{-3} \begin{bmatrix} 2 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$= \frac{1}{-3} \begin{bmatrix} 2(1) - 1(4) & 2(2) - 1(3) \\ -1(1) - 1(4) & -1(2) - 1(3) \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} A^{-1} \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{-3} \begin{bmatrix} -2 & 1 \\ -5 & -5 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} -2(-1) + 1(1) & -2(1) + 1(2) \\ -5(-1) + -5(1) & -5(1) - 5(2) \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & -15 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}$$

$$A^n = P D^n P^{-1}$$

$$A^4 = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 5^4 \end{bmatrix} \frac{1}{-3} \begin{bmatrix} 2 & -1 \\ -1 & -1 \end{bmatrix}$$

Quadratic Form / Quadratic expression
Homogeneous Expression: - An expression is called homogeneous expression, if each and every term of that expression must have same degree.

Quadratic Expression: - A homogeneous expression of second degree. In any number number of variables is called a quadratic expression & is represented by Q .

Note: -

* A quadratic expression Q in two variables x & y is taken as

$$Q \equiv ax^2 + 2hxy + by^2 \Rightarrow x^T A x$$

* A quadratic expression Q in three variables x, y, z is taken as

$$Q \equiv ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy.$$

$$\Downarrow \\ x^T A x$$

Note: - $\rightarrow$ 'A' is symmetric matrix. ~~this is~~

$\rightarrow$ for $Q \equiv x^T A x$

Find all eigen values of 'A'.

① Rank: - The no of non-zero eigen values of 'A' is called the rank of A.

② Index: - No of positive eigen values of 'A' is called Index of Q.

③ Signature:— The difference between the number of positive eigen value and no of negative eigen value of 'A' is called signature of 'Q'.

Nature:—

① Positive definite:—

If all the eigen values of 'A' are +ve.
It is positive definite.

② Negative definite:—

If all the eigen values of 'A' are -ve. It is negative definite.

③ Positive semidefinite:—

At least one eigen value zero remaining all are positive.

④ Negative semidefinite:—

At least one eigen value zero, remaining all are negative.

⑤ Indefinite

Some eigen values are positive, some eigen values are negative.

Canonical form / sum of squares

The canonical form of Quadratic form

$Q = x^T A x$ is taken as

$$= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2 \quad [\because \lambda_1, \lambda_2, \lambda_3 \text{ are eigen values of } A]$$

Q. Write canonical form of the Quadratic form $x^2 + 4y^2 + z^2 - 4xy + 2zx - 4yz$ and also find rank, index, signature & nature.

sol:- $Q \equiv x^2 + 4y^2 + z^2 - 4xy + 2zx - 4yz = \mathbf{x}^T \mathbf{A} \mathbf{x}$
 where $\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $\mathbf{x}^T = [x \ y \ z]$.

$$\mathbf{A} = \begin{matrix} & \begin{matrix} x & y & z \end{matrix} \\ \begin{matrix} x \\ y \\ z \end{matrix} & \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix} \end{matrix}$$

$$\mathbf{A} - \lambda \mathbf{I} = 0$$

$$\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = 0$$

$$\begin{bmatrix} 1-\lambda & -2 & 1 \\ -2 & 4-\lambda & -2 \\ 1 & -2 & 1-\lambda \end{bmatrix} = 0$$

$$| \mathbf{A} - \lambda \mathbf{I} | = 0$$

$$(1-\lambda)((4-\lambda)(1-\lambda) - 4) + 2(-2(1-\lambda) + 2) +$$

$$1(4 - 4 + \lambda) = 0$$

$$(1-\lambda)(4 - 4\lambda - \lambda + \lambda^2 - 4) + 2(-2 + 2\lambda + 2) + \lambda = 0$$

$$(1-\lambda)(\lambda^2 - 5\lambda) + 2(2\lambda) + \lambda = 0$$

$$\lambda^2 - 5\lambda - \lambda^3 + 5\lambda^2 + 4\lambda + \lambda = 0$$

$$-\lambda^3 + 6\lambda^2 = \lambda^3 - 6\lambda^2 = 0$$